



Geography: Global

# Tropical Forecast - Cone v3.0

## Document Version 1.1

Domain Portfolio: Tropical | Domain: Forecast | Usage Classification: Standard

Attribution Required: No

Attribution Requirements: N/A

### Overview

The Tropical Forecast - Cone API renders a cone representing the projected path of a tropical storm and current position information, based on 5 days of forecast advisory. For this API, JSON returned will always be GeoJSON.

Note: Storms will only pull from a designated source when calling by stormId.

- See the common usage and style guide which describes the common elements, error handling, language support, units and terminology used by the Weather Company Data APIs.
  - <https://ibm.co/APICom>

### HTTP Headers and Data Lifetime - Caching and Expiration

For details on appropriate header values as well as caching and expiration definitions, please see [The Weather Company Data | API Common Usage Guide](#).

### Translated Fields:

This TWC API handles the translation of phrases. However, when formatting a request URL a valid language must be passed along (see the language code table for the supported codes).

- |                      |                       |                          |
|----------------------|-----------------------|--------------------------|
| • stormType          | • stormSubType        | • stormDirectionCardinal |
| • latitudeHemisphere | • longitudeHemisphere | • directionCardinal      |

### URL Construction

| Atomic API URL Examples:                                                                                                                                                                                                                                                          | Aggregate Product Name | v3-tropical-cone |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|
| <b>Request by Source and Basin:</b> Required Parameters: source, basin, language, format, units, nautical, apiKey<br>https://api.weather.com/v3/tropical/cone?source=<source>&basin=<basin>&language=<language>&format=<format>&units=<units>&nautical=<nautical>&apiKey=<apiKey> |                        |                  |
| https://api.weather.com/v3/tropical/cone?source=default&basin=all&language=en-US&format=json&units=e&nautical=true&apiKey=yourApiKey                                                                                                                                              |                        |                  |
| <b>Request by Storm ID:</b> Required Parameters: stormId, language, format, units, nautical, apiKey<br>https://api.weather.com/v3/tropical/cone?stormId=<stormId>&language=<language>&format=<format>&units=<units>&nautical=<nautical>&apiKey=<apiKey>                           |                        |                  |
| https://api.weather.com/v3/tropical/cone?stormId=EP052016&language=en-US&format=json&units=e&nautical=true&apiKey=yourApiKey                                                                                                                                                      |                        |                  |

Valid Parameter Definitions

The valid parameter values listed below are unique to this API and are cases sensitive. These parameters support the ability to ‘filter’ the request by the supported sources and basins.

|                       |                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Parameter: source     |                                                                                                                                                                    |
| Valid Parameter Value | Description                                                                                                                                                        |
| default               | Data returned is limited to a TWC selected source for that basin. As additional sources are added worldwide, the default source for a particular basin may change. |
| all                   | Data returned for all sources for that basin.                                                                                                                      |
| Parameter: basin      |                                                                                                                                                                    |
| Valid Parameter Value | Description                                                                                                                                                        |
| all                   | Data returned for all supported basins (listed below).                                                                                                             |
| AL                    | Atlantic Ocean                                                                                                                                                     |
| CP                    | Central Pacific                                                                                                                                                    |
| EP                    | Eastern Pacific Ocean                                                                                                                                              |
| IO                    | Indian Ocean                                                                                                                                                       |
| SH                    | Southern Hemisphere                                                                                                                                                |
| WP                    | Western Pacific                                                                                                                                                    |

Nautical Parameter

If nautical=true, then any measurement that could be represented as knots is, regardless of the units parameter. Anything that is not subject to nautical form is left in the requested units (which is why you see pressures switch back and forth with units even when nautical=true)

|                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                    |
| units=e&nautical=true will give pressures in inches-of-mercury and speeds in knots<br>units=m&nautical=true will give pressures in millibars and speeds in knots<br>units=e&nautical=false will give pressures in inches-of-mercury and speeds in mph<br>units=m&nautical=false will give pressures in millibars and speeds in kph |

Data Elements & Definitions

| Field Name                       | Description                                                                       | Type    | Range                                                                                                              | Sample                                       | Nulls Allowed |
|----------------------------------|-----------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------|
| metadata                         | metadata object                                                                   | object  |                                                                                                                    |                                              | N             |
| next                             | Indicates if multiple queries are required                                        | number  |                                                                                                                    | 1346619600000                                | Y             |
| type                             | GeoJSON type attribute                                                            | string  | FeatureCollection                                                                                                  | "FeatureCollection",                         | N             |
| features                         | GeoJSON features array                                                            | [array] |                                                                                                                    |                                              | N             |
| properties                       |                                                                                   | object  |                                                                                                                    |                                              | N             |
| bulletinId                       | internal use only                                                                 | string  |                                                                                                                    | a94e1880a4870ae1b7d21049151d5e<br>dfa4e1011e | N             |
| stormKey                         | Unique storm identifier through life of the storm                                 | string  |                                                                                                                    | a87e1880a4870ae1b7d21049151d5e<br>dfa4e1011e | N             |
| stormId                          | Storm id. Format will vary by source. Not unique globally                         | string  |                                                                                                                    | EP022002                                     | N             |
| stormNumber                      | Given for only named storms leveraging JMA data. Otherwise null                   | integer |                                                                                                                    | null                                         | Y             |
| stormName                        | Name assigned by issuing agency. Named storms are in proper case                  | string  |                                                                                                                    | Alberto                                      | N             |
| alternateStormName               | 'Storm-name' populated in Japanese when data is fromJ MA. Otherwise will be'null' | string  |                                                                                                                    | null                                         | Y             |
| source                           | Issuing Agency                                                                    | string  | NHC, WPC, CPHC, JTWC                                                                                               | NHC                                          | N             |
| designatedSource                 | true if a TWC selected provider                                                   | boolean | true or false                                                                                                      | TRUE                                         | N             |
| basin                            | Basin for storm                                                                   | string  | See Valid Parameter Definitions above                                                                              | WP                                           | N             |
| issueOffice                      | office issuing bulletin                                                           | string  |                                                                                                                    | KNHC                                         | N             |
| wmold                            | WMO identifier                                                                    | string  |                                                                                                                    | WTNT21                                       | Y             |
| issueDateTime                    | Time Issuing Agency creates the bulletin                                          | ISO     | ISO 8601 – YYYY-MM-DDTHH:MI:SS±HHMI or Z                                                                           | 2015-01-13T01:00:00-00:00                    | N             |
| issueDateTimeZoneCode            | IANA code for issueDateTime                                                       | string  |                                                                                                                    | America/New_York                             | N             |
| issueDateTimeZoneAbbreviation    | Time zone abbreviation for issueDateTime                                          | string  |                                                                                                                    | EDT                                          | N             |
| advisoryNumber                   | Issuing Agency advisory number                                                    | string  |                                                                                                                    | 12A                                          | N             |
| advisoryDateTime                 | Advisory time                                                                     | ISO     | ISO 8601 – YYYY-MM-DDTHH:MI:SS±HHMI or Z                                                                           | 2015-01-13T10:00:15-08:00                    | N             |
| advisoryDateTimeZoneCode         | IANA code for advisoryDateTime                                                    | string  |                                                                                                                    | America/New_York                             | N             |
| advisoryDateTimeZoneAbbreviation | Time zone abbreviation for issueDateTime                                          | string  |                                                                                                                    | EDT                                          | N             |
| currentPosition                  |                                                                                   | object  |                                                                                                                    |                                              | N             |
| latitude                         | Current latitude of the storm                                                     | decimal | Negative if in the Southern Hemisphere                                                                             | 42.5                                         | N             |
| latitudeHemisphere               | Latitude hemisphere                                                               | string  | N,S                                                                                                                | N                                            | N             |
| longitude                        | Current longitude of the storm                                                    | decimal | Negative if in the Western Hemisphere                                                                              | -59.8                                        | N             |
| longitudeHemisphere              | Longitude hemisphere                                                              | string  | E,W                                                                                                                | W                                            | N             |
| stormTypeCode                    | Encoded storm type                                                                | string  | TD, TS, HU, SS, PT, RO, SD, EX, LOW, PTC, TY, ST, TC,                                                              | HU                                           | N             |
| stormType                        | Storm type                                                                        | string  | Tropical Depression, Tropical Storm, Hurricane, Subtropical Storm, Post-tropical cyclone, Remnants of, Subtropical | Hurricane                                    | N             |

|                        |                                                        |         |                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                        |   |
|------------------------|--------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                        |                                                        |         | Depression, Extratropical, Extra Tropical Low, Potential Tropical Cyclone, Typhoon, Super Typhoon, Tropical Cyclone,                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                        |   |
| stormSubTypeCode       | Encoded storm sub type                                 | string  | 1, 2, 3, 4, 5                                                                                                                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                      | Y |
| stormSubType           | Storm sub type                                         | string  | Category 1 Hurricane<br>Category 2 Hurricane<br>Category 3 Hurricane<br>Category 4 Hurricane<br>Category 5 Hurricane<br>Category 1 Tropical Cyclone<br>Category 2 Tropical Cyclone<br>Category 3 Tropical Cyclone<br>Category 4 Tropical Cyclone<br>Category 5 Tropical Cyclone<br>Cyclonic Storm<br>Severe Cyclonic Storm<br>Very Severe Cyclonic Storm<br>Extremely Severe Cyclonic Storm<br>Super Cyclonic Storm | Category 1 Hurricane<br><br>Note:<br>In EP, CP, and AL, only stormType=HU (Hurricane) has a subType<br>In WP, no stormType has a subType<br>In IO and SH, stormType=TC (Tropical Cyclone) has a subType. If stormType=TC in another basin, it will not have a subType. | Y |
| headline               | List of headlines. [ list ]                            | [array] | For NHC data headline is US English only and will be populated when available<br>For JTWC the data headline will be null.                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                        | Y |
| headline               | Headline extracted from bulletin                       | string  |                                                                                                                                                                                                                                                                                                                                                                                                                     | HURRICANE CONDITIONS EXPECTED TODAY                                                                                                                                                                                                                                    | N |
| maximumSustainedWind   | Maximum sustained winds                                | integer |                                                                                                                                                                                                                                                                                                                                                                                                                     | 45                                                                                                                                                                                                                                                                     | N |
| windGust               | Wind gusts                                             | integer |                                                                                                                                                                                                                                                                                                                                                                                                                     | 70                                                                                                                                                                                                                                                                     | Y |
| minimumPressure        | Minimum central pressure                               | decimal |                                                                                                                                                                                                                                                                                                                                                                                                                     | 29.5                                                                                                                                                                                                                                                                   | Y |
| heading                | Heading information for the storm                      | object  |                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                        | N |
| stormDirection         | Current direction of the storm in degrees              | string  | 0 to 360 or stationary or drifting, meandering                                                                                                                                                                                                                                                                                                                                                                      | 340                                                                                                                                                                                                                                                                    | N |
| stormDirectionCardinal | Cardinal wind direction                                | string  | N, S,E,W, NNE, NE,ENE,ESE, SE, SSE, SSW, SW, WSW, WNW, NW,NNW, Almost Stationary, Stationary, Drifting , Meandering                                                                                                                                                                                                                                                                                                 | NNW                                                                                                                                                                                                                                                                    | N |
| stormSpeed             | Forward speed of the storm                             | integer |                                                                                                                                                                                                                                                                                                                                                                                                                     | 12                                                                                                                                                                                                                                                                     | Y |
| nearbyLocation         | Geographical references for a storm. [list of objects] | object  | Only populated from TCP and TCU bulletin.<br>For JTWC the data headline will be null.                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                        | Y |
| locationName           | Location name                                          | string  | Always uppercase                                                                                                                                                                                                                                                                                                                                                                                                    | CAPE HATTERAS NORTH CAROLINA                                                                                                                                                                                                                                           | N |
| distance               | Distance from the center of the storm                  | integer |                                                                                                                                                                                                                                                                                                                                                                                                                     | 240                                                                                                                                                                                                                                                                    | Y |
| directionCardinal      | Direction of locationName from the center of the storm | string  | N, S,E,W, NNE, NE,ENE,ESE, SE, SSE, SSW, SW, WSW, WNW, NW,NNW, Almost Stationary, Stationary, Drifting , Meandering                                                                                                                                                                                                                                                                                                 | N                                                                                                                                                                                                                                                                      | Y |
| cityName               | Nearest city                                           | string  | Only for CONUS. Always in proper case                                                                                                                                                                                                                                                                                                                                                                               | Cape Hatteras                                                                                                                                                                                                                                                          | Y |
| stateCode              | Nearest state                                          | string  | Only for CONUS. Always in uppercase                                                                                                                                                                                                                                                                                                                                                                                 | NC                                                                                                                                                                                                                                                                     | Y |
| windRadii              | Storm wind radius values. [list of objects]            | object  |                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                        | Y |
| radiiWindSpeed         | Wind speed for the associated quadrant                 | integer |                                                                                                                                                                                                                                                                                                                                                                                                                     | 64                                                                                                                                                                                                                                                                     | N |

|             |                                                                                                   |         |         |           |   |
|-------------|---------------------------------------------------------------------------------------------------|---------|---------|-----------|---|
| NE          | Distance from the center of the storm in the NE quadrant where radii_wspd begin                   | integer |         | 75        | N |
| SE          | Distance from the center of the storm in the SE quadrant where radii_wspd begin                   | integer |         | 75        | N |
| SW          | Distance from the center of the storm in the SW quadrant where radii_wspd begin                   | integer |         | 80        | N |
| NW          | Distance from the center of the storm in the NW quadrant where radii_wspd begin                   | integer |         | 80        | N |
| seaRadii    | Wave Height Radius Values Object                                                                  | object  |         |           | Y |
| NE          | Distance from the center of the storm in the NE quadrant where wave_height values begin           | integer |         | 12        | N |
| SE          | Distance from the center of the storm in the SE quadrant where wave_height values begin           | integer |         | 75        | N |
| SW          | Distance from the center of the storm in the SW quadrant where wave_height values begin           | integer |         | 75        | N |
| NW          | Distance from the center of the storm in the NW quadrant where wave_height values begin           | integer |         | 80        | N |
| type        | GeoJSON type                                                                                      | string  | Feature | "Feature" | N |
| geometry    | GeoJSON geometry                                                                                  | object  |         |           | N |
| type        | GeoJSON type                                                                                      | string  | Polygon | "Polygon" | N |
| coordinates | Coordinates array                                                                                 | [array] |         |           | N |
| coordinates | Array of GeoJSON coordinates ordered as: (longitude, latitude) which defines the polygon geometry | [array] |         |           | N |

JSON Sample

```
// Response Collapsed for Presentation Purposes
{
  "metadata": {
    "next": null
  },
  "type": "FeatureCollection",
  "features": [
    {
      "properties": {
        "bulletinId": "d978a4a7cdb30852a517ef7ec741f05d",
        "stormKey": "f4b8aa1e70f9b4866f061703b6f298fd",
        "stormId": "AL112017",
        "stormNumber": null,
        "stormName": "Irma",
        "alternateStormName": null,
        "source": "NHC",
        "designatedSource": true,
        "basin": "AL",
        "issueOffice": "KNHC",
        "wmold": "WTNT21",
        "issueDateTime": "2018-04-19T20:55:00Z",
```

```
"issueDateTimeZoneCode": "Etc/UTC",
"issueDateTimeZoneAbbreviation": "UTC",
"advisoryNumber": "39",
"advisoryDateTime": "2018-04-19T17:00:00-04:00",
"advisoryDateTimeZoneCode": "America/New_York",
"advisoryDateTimeZoneAbbreviation": "EDT",
"currentPosition": {
  "latitude": 22.1,
  "latitudeHemisphere": "N",
  "longitude": -76.5,
  "longitudeHemisphere": "W",
  "stormTypeCode": "HU",
  "stormType": "Hurricane",
  "stormSubTypeCode": "4",
  "stormSubType": "Category 4 Hurricane",
  "headline": null,
  "minimumPressure": 925,
  "maximumSustainedWind": 250,
  "windGust": 305,
  "heading": {
    "stormDirection": "280",
    "stormDirectionCardinal": "W",
    "stormSpeed": 19
  },
  "nearbyLocation": [],
  "windRadii": [
    {
      "radiiWindSpeed": 63,
      "NE": 296,
      "SE": 222,
      "SW": 148,
      "NW": 278
    },
    {
      "radiiWindSpeed": 93,
      "NE": 185,
      "SE": 167,
      "SW": 93,
      "NW": 148
    },
    {
      "radiiWindSpeed": 119,
      "NE": 111,
      "SE": 111,
```

```
    "SW": 56,  
    "NW": 111  
  },  
],  
  "seaRadii": {  
    "waveHeight": 4,  
    "NE": 944,  
    "SE": 333,  
    "SW": 111,  
    "NW": 888  
  }  
}  
},  
"type": "Feature",  
"geometry": {  
  "type": "Polygon",  
  "coordinates": [  
    [  
      [-76.3447,  
        22.1014  
    ],  
    [  
      [-76.3474,  
        22.0731  
    ],  
    [  
      [-76.356,  
        22.0459  
    ],  
    [  
      [-76.3701,  
        22.0207  
    ],  
    [  
      [-76.3479,  
        22.1296  
    ],  
    [  
      [-76.3447,  
        22.1014  
    ]  
  ]  
]  
]
```

|                                                             |  |
|-------------------------------------------------------------|--|
| <div><div>}</div><div>}</div><div>]</div><div>}</div></div> |  |
|                                                             |  |